

PHASE 7 – PROJECT DOCUMENTATION

1. INTRODUCTION

The Project Documentation phase presents the complete documentation of the **Script-Controlled ACL – Restrict Record Access Based on Field Value** project. This phase consolidates the requirements, system design, implementation activities, testing results, security configuration, and project outcomes into a structured document.

The project was developed in ServiceNow to control access to records in the **Institution Details** table based on user roles and the Branch field. The configured ACLs provide different permissions for reading, creating, updating, and deleting records.

2. PROJECT TITLE

Script-Controlled ACL – Restrict Record Access Based on Field Value

3. PROJECT OVERVIEW

The project implements role-based and condition-based access control in ServiceNow.

The main requirement is to restrict access to Institution Details records according to the **Branch** value. Users with the required bb1 role can access records satisfying the configured **Branch = EEE** condition, while administrators retain full access.

Additional ACLs are configured to control Create, Write, and Delete operations using separate roles.

4. PROBLEM STATEMENT

In an application containing student and institution information, all users should not have unrestricted access to every record.

Without proper access control, unauthorized users may be able to view, create, modify, or delete records. Therefore, a security mechanism is required to restrict record access based on user roles and record conditions.

This project addresses the requirement by implementing ACLs in ServiceNow.

5. PROJECT OBJECTIVES

The main objectives of the project are:

- To create a custom Institution Details table.
- To create an EEE test user.
- To create and assign custom roles.
- To implement a script-controlled Read ACL.
- To restrict records according to the Branch condition.
- To control Create, Write, and Delete operations.
- To provide administrators with full access.
- To verify authorized and unauthorized access.
- To document the implementation and testing results.

6. REQUIREMENT ANALYSIS

6.1 User Requirement

An EEE test user is created with the following details:

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

6.2 Role Requirements

The following custom roles are created:

- bb1
- bb2
- bb3
- bb4

The roles are assigned according to the required access operations.

6.3 Table Requirements

The custom table is:

Institution Details

Table name:

u_institution_details

The table contains student and institution-related information.

6.4 Field Requirements

Field	Type
Student Roll Number	Auto Number
Student Name	Reference - User
Faculty Name	Reference - User
Branch	Choice
Email	String
Phone Number	String
Description	Multi String

The Branch choice values are:

- ECE
- EEE
- CSE

7. SYSTEM DESIGN

The system follows a role-based and condition-based access-control model.

7.1 Access Control Flow

User

|

v

ServiceNow

|

v

User Role Verification

|

+---- bb1 ----> Read Access

|

+---- bb2 ----> Create Access

|

+---- bb3 ----> Write Access

|

+---- bb4 ----> Delete Access

|

+---- admin --> Full Access

|

v

ACL Evaluation

|

v

Institution Details Records

For the Read operation, the ACL also contains the configured **Branch = EEE** data condition.

8. PROJECT IMPLEMENTATION

8.1 User Creation

An EEE test user was created in ServiceNow under User Administration → Users.

The user was used for testing the configured access-control rules.

8.2 Role Creation

Four custom roles were created:

- bb1 for Read access
- bb2 for Create access
- bb3 for Write access
- bb4 for Delete access

The required roles were assigned to the EEE test user.

8.3 Institution Details Table

The **Institution Details** table was created under System Definition → Tables.

The table was configured with the required student, faculty, branch, email, phone, and description fields.

8.4 Test Record Creation

Multiple records were created with different Branch values such as:

- EEE
- ECE
- CSE

These records were used to verify branch-based record access.

9. ACL CONFIGURATION

9.1 Read ACL

A Record-level Read ACL was created for:

u_institution_details

The ACL contains:

- Operation: Read
- Active: True
- Advanced: True
- Required role: bb1
- Data condition: **Branch is EEE**

The script allows administrators and users with the required bb1 role, while denying other users.

```
(function () {  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    // Allow users with the required role  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    // Deny access for all others  
    return false;  
})();
```

The Branch restriction is configured through the **Data condition: Branch is EEE**.

9.2 Create ACL

A Record-level Create ACL was configured with:

- Operation: Create
- Required role: bb2
- Active: True

This allows users with the required Create permission to create records.

9.3 Write ACL

A Record-level Write ACL was configured with:

- Operation: Write
- Required role: bb3
- Active: True

This allows authorized users to modify existing records.

9.4 Delete ACL

A Record-level Delete ACL was configured with:

- Operation: Delete
- Required role: bb4
- Active: True

This allows authorized users to delete records.

10. ACCESS CONTROL MODEL

Operation	Required Role	Access
Read	bb1	EEE records according to the configured condition
Create	bb2	Create records
Write	bb3	Modify records
Delete	bb4	Delete records
Administrator	admin	Full access

11. PROJECT TESTING

Testing was performed using different users and role combinations.

The following test cases were used:

Test Case	User/Role	Action	Expected Result
TC01	bb1	Open records	Permitted EEE records are accessible
TC02	No required role	Open records	Access is restricted
TC03	bb1 + bb2	Create record	New record can be created
TC04	bb1 + bb2 + bb3	Edit record	Record can be modified
TC05	bb1 + bb2 + bb3 + bb4	Delete record	Record can be deleted
TC06	Administrator	Open records	Full access is available

12. TEST RESULTS

The implemented ACL configuration was tested successfully.

Test Case	Actual Result	Status
TC01	EEE User accessed the EEE record successfully	Pass
TC02	User without required role received a security constraint message	Pass
TC03	New button was available and record creation was permitted	Pass
TC04	Update button was available and record was editable	Pass
TC05	EEE record was deleted successfully	Pass
TC06	Administrator viewed records from ECE, EEE, and CSE branches	Pass

Therefore, all documented test cases produced the expected results.

13. SCREENSHOT AND EVIDENCE DOCUMENTATION

The project implementation and testing were supported with screenshots.

Development Evidence

The following implementation screenshots were captured:

The screenshot displays the ServiceNow User Administration interface for a user named 'EEE User'. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The user profile is shown with the following details:

- User ID:** EEE User
- First name:** EEE
- Last name:** User
- Title:** (empty)
- Department:** (empty)
- Email:** eecuser@gmail.com
- Language:** -- None --
- Calendar integration:** Outlook
- Time zone:** System (America/Los_Angeles)
- Date format:** System (2026-08-19)
- Business phone:** (empty)
- Mobile phone:** (empty)
- Photo:** Click to add...

Additional settings and actions are visible:

- Password needs reset:** ☐
- Locked out:** ☐
- Active:** ☒
- Identity type:** Human
- Internal Integration User:** ☐
- Buttons:** Update, Set Password, Delete
- Related Links:** View linked accounts, View Subscriptions, Reset a password
- Roles (5):** Groups (1), Delegates, User Client Certificates
- Footer:** User = FFF User

Figure 7.1 – EEE User Creation

Name	Description	Elevated privilege
bb1	Brief role purpose	false
bb2	Brief role purpose	false
bb3	Brief role purpose	false
bb4	Brief role purpose	false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.	false
caching_admin	Role for monitoring and configuring caching framework related tables	false
cannot_read_suggestions	Users with this role are not authorized to use suggestion service.	false
canvas_admin	Admin role for Canvas Core Application	false

Figure 7.2 – Custom Role Creation

User - EEE User

Locked out ☐ Active ☒ Identity type: Human Internal Integration User ☐

Mobile phone: Photo: Click to add...

Update Set Password Delete

Related Links
[View linked accounts](#)
[View Subscriptions](#)
[Reset a password](#)

Roles (5) Groups (1) Delegates User Client Certificates

Role	State	Inherited	Inheritance Count
bb1	Active	false	
bb2	Active	false	
bb3	Active	false	
snc_required_script_writer_permission	Active	true	1
bb4	Active	false	

Figure 7.3 – Role Assignment

Table Institution Details

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Institution Details

* Name u_institution_details

Application Global

Remote Table

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false
Created by	String	(empty)		40	false
Branch	Choice	(empty)		40	false
Student Name	Reference	User		32	false
Created	Date/Time	(empty)		40	false
Faculty Name	Reference	User		32	false
Description	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Student Roll Number	String	(empty)		40	false

Figure 7.4 – Institution Details Table

servicenow All Favorites History Workspaces Admin Table - Institution Details

Search

Table Institution Details

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false
Created by	String	(empty)		40	false
Branch	Choice	(empty)		40	false
Student Name	Reference	User		32	false
Created	Date/Time	(empty)		40	false
Faculty Name	Reference	User		32	false
Description	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Student Roll Number	String	(empty)		40	false
Updated by	String	(empty)		40	false
Phone Number	String	(empty)		40	false
Email	String	(empty)		40	false
Updated	Date/Time	(empty)		40	false
+ Insert a new row...					

Delete Update Delete All Records

Show desktop

Figure 7.5 – Table Fields

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
Branch	Choice	(empty)	40		false
Student Name	Reference	User	32		false
Created	Date/Time	(empty)	40		false
Faculty Name	Reference	User	32		false
Description	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Student Roll Number	String	(empty)	40		false
Updated by	String	(empty)	40		false
Phone Number	String	(empty)	40		false
Email	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
+ Insert a new row...					

Figure 7.6 – Sample Records

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
 * Operation: read
 Application: Global
 Active: ☒
 Advanced: ☒
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Institution Details [u_institution_details]
 Description:
 Conditions:
 Requires role:
 Role:
 + Insert a new row...

Figure 7.7 – ACL

servicenow All Favorites History Workspaces Admin Access Control - New Record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: create Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Insert a new row...

Figure 7.8 – Create ACL Read

servicenow All Favorites History Workspaces Admin Access Control - New Record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: write Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Insert a new row...

Figure 7.9 – Write ACL

servicenow All Favorites History Workspaces Admin **Access Control - New Record** Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: delete Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Insert a new row...

Figure 7.10 – Delete ACL

servicenow All Favorites History Process Mining Workspace **Institution Details** Search Actions on selected rows... New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Abraham Lincoln		System Administrator	IN0002

1 to 1 of 1

Figure 7.11 – User Access Verification

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Abraham Lincoln		Adela Cervantsz	IN0003
(empty)			(empty)		(empty)	
ECE			Abel Tuter		Adela Cervantsz	IN00001
EEE			Abraham Lincoln		System Administrator	IN0002

Figure 7.12 – Administrator Access Verification

Testing Evidence

The testing screenshots are documented as:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Abraham Lincoln		System Administrator	IN0002

Figure 7.13 – EEE User Access

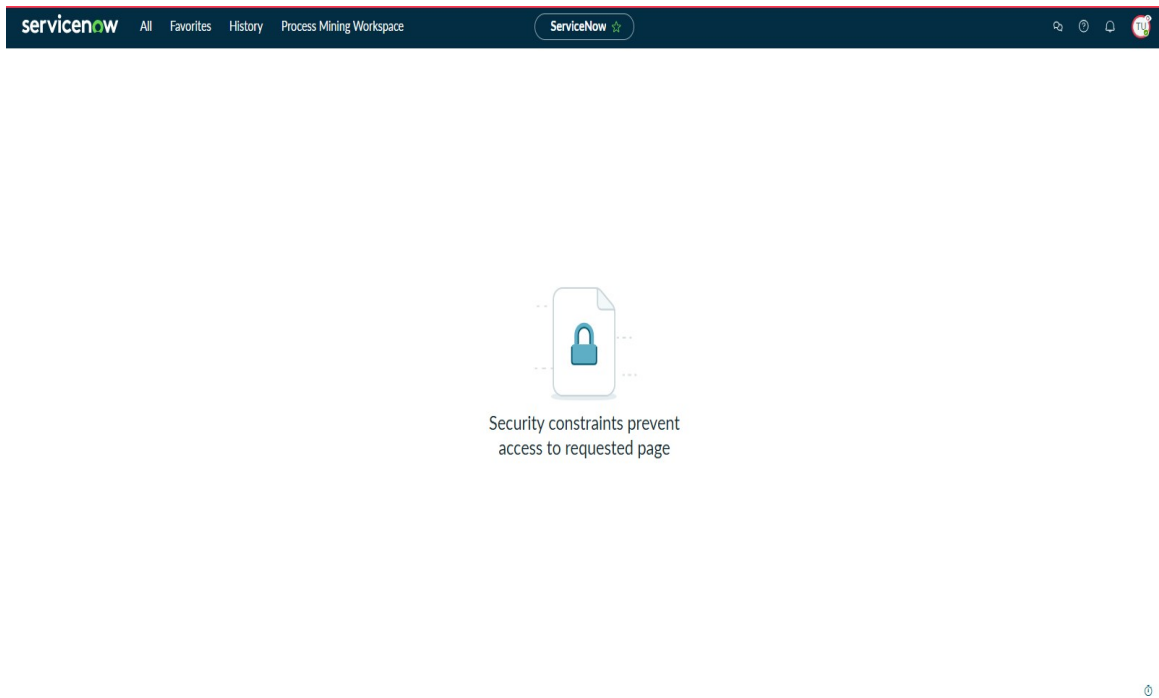


Figure 7.14 – Restricted User Access

The screenshot shows a ServiceNow table view for 'Institution Details'. The table has columns for Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. A single record is displayed for the 'EEE' branch, with the Faculty Name 'Abraham Lincoln', Student Name 'System Administrator', and Student Roll Number 'IN0002'. The table includes a search bar, a filter dropdown, and a 'New' button. The bottom of the page shows a pagination bar indicating '1 to 1 of 1' records.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Abraham Lincoln		System Administrator	IN0002

Figure 7.15 – Create Operation

The screenshot shows the 'Institution Details' form in ServiceNow. The header includes the ServiceNow logo, navigation tabs (All, Favorites, History, Process Mining Workspace), and a breadcrumb trail 'Institution Details - Created 2026-09-23 07:55:03'. The form fields are as follows:

Student Name	Alejandra Prenatt	Description	
Faculty Name	Abraham Lincoln	Phone Number	
Student Roll Number	IN0002	Email	
Branch	EEE		

Buttons for 'Update' and 'Delete' are located at the bottom left of the form.

Figure 7.16 – Write Operation

The screenshot shows the 'Institution Details' list view in ServiceNow. The header includes the ServiceNow logo, navigation tabs (All, Favorites, History, Process Mining Workspace), and a breadcrumb trail 'Institution Details'. The table has the following columns:

	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
 No records to display							

Figure 7.17 – Delete Operation

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Abraham Lincoln		Adela Cervantsz	IN0003
EEE			Abel Tuter		Adela Cervantsz	
(empty)			(empty)		(empty)	
ECE			Abel Tuter		Adela Cervantsz	IN00001

Figure 7.18 – Administrator Access

These screenshots provide evidence of the implemented configuration and testing activities.

14. SECURITY IMPLEMENTATION

Security was implemented using ServiceNow Access Control Lists.

The Read ACL combines:

- User role verification
- Branch-based data condition
- Administrator access

The Create, Write, and Delete ACLs provide separate permissions through the corresponding roles.

This approach prevents users without the required authorization from performing protected operations.

15. PROJECT OUTCOME

The project successfully implemented a script-controlled ACL mechanism in ServiceNow.

The final system provides:

- Branch-based Read restriction.
- Role-based Create permission.
- Role-based Write permission.
- Role-based Delete permission.

- Administrator full access.
- Unauthorized access restriction.
- Verified access-control behavior through testing.

All six documented test cases were completed with **Pass** status.

16. CHALLENGES ENCOUNTERED

During implementation, access control had to be configured carefully because multiple ACL operations and roles were involved.

The main areas requiring attention were:

- Creating and assigning the correct roles.
- Configuring the Branch-based Read condition.
- Applying separate roles for Create, Write, and Delete operations.
- Testing unauthorized access.
- Verifying administrator access.
- Confirming that the ACL configuration behaved as expected.

17. FUTURE ENHANCEMENTS

The project can be enhanced in the future by:

- Adding more branches and department-based conditions.
- Introducing additional user roles.
- Creating more complex access conditions.
- Adding dynamic access rules based on user attributes.
- Extending the ACL logic to other custom tables.
- Adding more automated testing scenarios.
- Providing a more detailed access-control dashboard.

18. CONCLUSION

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project demonstrates how ServiceNow ACLs can be used to implement controlled access to records.

The Institution Details table was created with the required fields and sample records. Custom roles were created for different operations, and ACLs were configured for Read, Create, Write, and Delete permissions.

The Read ACL applies the configured **Branch = EEE** condition, while administrators retain full access. The project was tested using authorized, unauthorized, and administrator users, and all documented test cases achieved a **Pass** result.

Thus, the project documentation provides a complete record of the requirements, design, implementation, testing, evidence, and outcome of the ServiceNow ACL implementation.